

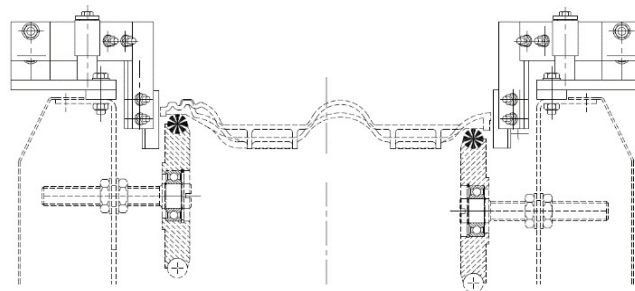
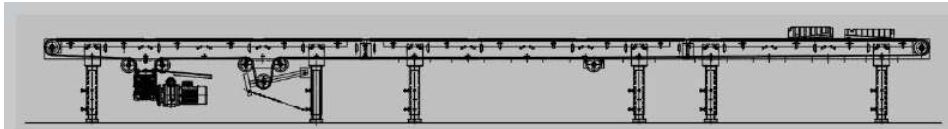
TRANSPORT LINE OF TILES AND MOULDS-Relationship SOP

MACHINE TYPE:TM103 SERIAL NUMBER :M.02208

FP McCANN (Cadeby)

ORDER : C22004

YEAR OF PRODUCTION 2023



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- SOP M.02209.01 Transport line of tiles and moulds.
- SOP M.02209.01A Polycord belt, Pallet guide 20x12x400, Guide for polycord 20, Pulleys 250 / 148 / 100
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TRANSPORT LINE OF TILES AND MOULDS

MACHINE TYPE: TM103 SERIAL NUMBER :M.02209

Transport line of tiles and moulds.

General Description and Function
MOULD + FRESH TILE CONVEYORS

*Each conveyor must be clearly identified and labeled to enable proper tracking of all preventive and corrective maintenance activities.

Critical Points and Risk Management

*The following conveyors require the most stringent monitoring due to specific contamination risks.

Extruder to Racker:

*High risk of oil and concrete contamination due to potential over-application of mould release agent and concrete spillage from the mould sides at the extruder outlet.

SUPERVISION LEVEL

DAILY

17055 POLYCORD BELT (400 m)

252664 PALLET GUIDE 20x12x400 (5 units)

Visual inspection of key wear components.

WEEKLY

19498 BEARING (PULLEYS 178 u)

3422 DRIVING PULLEYS 250 (30 units)

Cleaning and functional Check of drive elements

MONTHLY

289642 GEARMOTOR (14 units)

296298 GEARMOTOR (1 unit)

Comprehensive inspection and verification.

DAILY

*Polycord alignment guides

*General condition of the polycord (no major visible damage)

*Rotation and condition of driven pulleys (d 148) and drive pulleys (d250)

WEEKLY

Polycord Tension:

*Ensure it is within the specified range.

Guide Wear:

*Check for excessive wear on polycord guides.

Polycord Integrity:

*Inspect for reduced diameter, cuts, or surface erosion.

Pulleys & Bearings:

Clean dirt and debris.

*Check for wear.

*Perform a bearing play test on driven

Area Inspection:

*Look for oil Oil contamination on floor in the gear box area.

CLEANING

WEEKLY

*Thoroughly clean all specified components to prevent performance degradation:

Components:

*Polycord, guides, and pulleys.

Method:

*Use appropriate tools (scrapers, brushes, dry cloths) to remove all concrete dust, oil residue, and polycord debris from surfaces.

Special Attention:

*Pay particular attention to the assembly-side polycord, guides, and pulleys, as these are most critical for product placement.

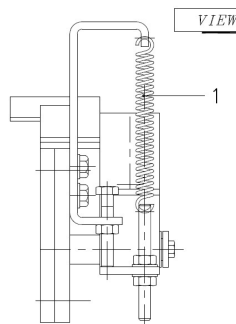
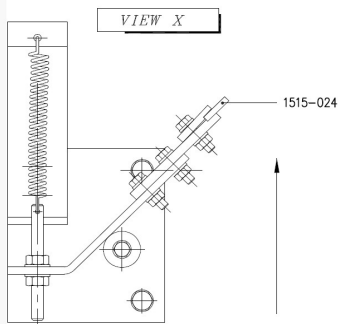
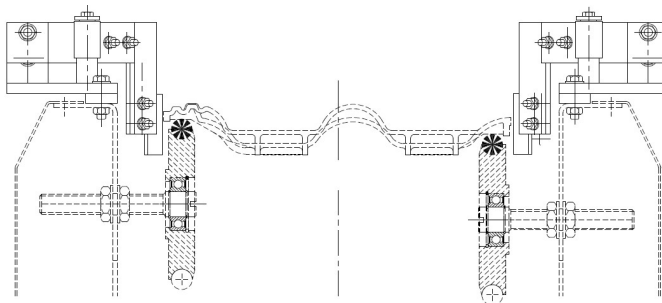


VORTEX HYDRA	STANDARD OPERATING PROCEDURE	Code: SOP M.02209.01A Date: 15/12/25 Page: 3 de 6
TRANSPORT LINE OF TILES AND MOULDS MACHINE TYPE:TM103 SERIAL NUMBER :M.02209		
Polycord belt, Pallet guide 20x12x400, Guide for polycord 20, Pulleys 250 / 148 / 100		
17055 POLYCORD BELT POLY-20 46640 GUIDE FOR POLYCORD 20 3422 DRIVING PULLEY 250 / POLYCORD 20 8233 CHAMFERED PULLEY 148 / POLYCORD 20 8234 SNUB PULLEY 148 / POLYCORD 20	Criteria for Determining Wear and Replacement Need 1. For the POLYCORD (e.g., Ø 20 mm) The polycord is worn and must be changed when it exhibits one or more of the following defects: *Diameter Reduction: The diameter measured at any point is ≤ 18 mm (a loss of 10% or more of the nominal diameter). *Cut or Laceration Damage: Presence of deep cuts that expose the internal core or compromise structural integrity. *Irregular Wear (Erosion): Surface with deep grooves, localized flattening, or an oval shape that compromises uniform contact with guides and pulleys. *Loss of Elastic Properties: The cord is stiff, shows oxidation cracks (crazing), or has lost its nominal tension and elasticity. Verdict: "The polycord exhibits critical wear. It must be replaced." 2. For the GUIDES (Round or Angular) The guide is worn and must be changed when one or more of the following defects are observed: *Pronounced Groove or Wear Track: Presence of a visible and palpable groove (channel) in the polycord's contact and running area. It is considered critical if the groove depth exceeds 1 mm. Sharp Edges or Burrs: The sliding surface has developed sharp edges or irregularities (burrs) that could damage the polycord. *Deformation: The guide is bent, twisted, or has lost its original geometry (straight or with the designed curvature). Excessive Play: The polycord has abnormal lateral movement or "play" within the guide, indicating wear that has unacceptably increased the clearance. Verdict: "The guide exhibits critical wear. It must be replaced." 	
		Wear Criterion (When to 1. Diameter ≤ 18 mm. 2. Deep cuts or lacerations. 3. Oval shape or deep grooves. 4. Stiffness and cracking. Verification: Visual / Tactile / Feeler gauge Wear Criterion (When to 1. Groove depth > 1 mm. 2. Presence of sharp edges or burrs 3. Visible deformation. 4. Excessive polycord play. Verification: Visual / Tactile / Feeler gauge

TRANSPORT LINE OF TILES AND MOULDS

MACHINE TYPE:TM103 SERIAL NUMBER :M.02209

Pallet Scrapping Device



General Description and Function

*This device consists of two side scrapers or plates, located on each side of the mould in the extrusion line. Its function is to continuously and automatically clean concrete residues adhering to the two sides of the mould.

*Effective cleaning is essential to ensure the mould travels correctly inside the extruder, maintaining the product's dimensional quality and protecting the equipment.

Critical Points and Risk Management

Inadequate Mould Cleaning:

*Insufficient cleaning of the mould sides is the primary risk.

*Accumulated residue causes excessive wear on the extruder's side plates and, consequently, can lead to tile deformation as it exits the die head, resulting in product rejection.

Wear of Scraper Components:

*The scraper plates and their tension springs are subject to continuous wear from friction and exposure to concrete.

*Their optimal condition is critical to maintaining the correct cleaning pressure and angle.

SUPERVISION LEVEL

DAILY

477 PLATE (2 units)

Inspect wear on the contact edge.

Verify it is secure and correctly aligned with the mould

WEEKLY

19063 Tension Spring (2 units)

CLEANING

DAILY

*The work area under the scrapers can accumulate generated waste, interfering with the movement of the mould or the device itself if not cleaned regularly.

TRANSPORT LINE OF TILES AND MOULDS
MACHINE TYPE:TM103 SERIAL NUMBER :M.02209
Tiles Blowing



General Description and Function

*This equipment consists of a high-capacity centrifugal fan designed to generate a powerful, directed airflow. Its function is to remove loose debris (dust, grit, particles) from the surfaces of tiles or molds passing beneath its line of action by air drag, ensuring product cleanliness before the subsequent stages of the process.

Critical Points and Risk Management

Airflow Effectiveness:

*Air pressure and flow rate are the critical parameters. A reduction in fan performance leads to ineffective cleaning, leaving debris on the tiles that can affect surface quality or downstream operations.

Wear and Imbalance:

*The fan blades and motor bearings are subject to wear. Imbalance or damage to the blades causes excessive vibration, abnormal noise, and ultimately catastrophic failure of the motor or fan structure.

Clogging from Dirt:

*The intake grille and the inside of the housing can become clogged with dust drawn from the environment or generated by the process itself, reducing airflow and straining the motor.

SUPERVISION LEVEL

MONTHLY

271695 CENTRIFUGAL FAN (1 u)

Operational Verification:

*Check that the air system delivers the intended flow and pressure for effective cleaning.

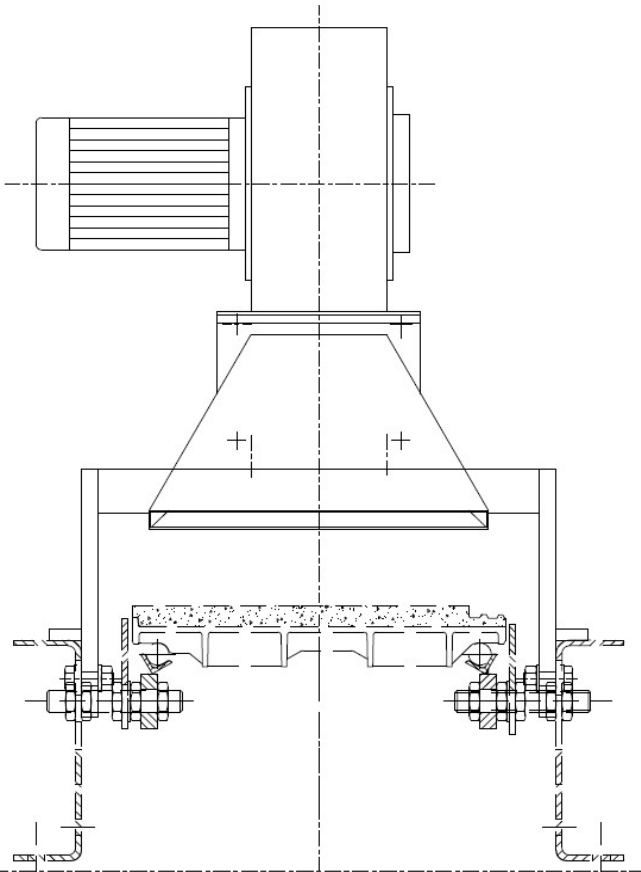
Noise and Vibration Check:

*Verify that the motor and blades do not emit abnormal noises or vibrations indicating wear or misalignment.

CLEANING

WEEKLY

*The exterior surfaces of the equipment and the intake grille to remove dust and debris that may affect its operation.



VORTEX HYDRA	STANDARD OPERATING PROCEDURE	Code: SOP M.02209.04 Date: 15/12/25 Page: 6 de 6
TRANSPORT LINE OF TILES AND MOULDS		
MACHINE TYPE:TM103 SERIAL NUMBER :M.02209		
Pallets Overlapping Safety Device TLE Profile		
		General Description and Function *This safety device consists of a protective plate (TLE Profile) mounted on a pivot shaft. Its function is to physically detect the incorrect overlapping or stacking of moulds or pallets on the transport line. The plate must have sufficient clearance to allow a single mould to pass underneath normally. If one mould drags or pushes the previous one (overlapping), it lifts and tilts the plate forward. This movement is detected by an inductive proximity switch installed in the assembly, which immediately triggers an alarm or an emergency line stop to prevent damage to equipment. Critical Points and Risk Management <u>Safety Sensor Failures:</u> *The device's primary function depends entirely on the correct operation of the inductive sensor. *A failure (damaged, misaligned, or dirty sensor occur which can cause serious jams, mould damage, and production stoppages.
		SUPERVISION LEVEL DAILY <u>Sensor Functional Test:</u> *Verify the inductive alarm detector is working correctly by manually moving the Plate-TLE Profile to simulate an overlap and confirm the alarm signal activates. WEEKLY 6663 Shaft (1 u) 50509 Strap -SEA WAVE (1 u) 252094 Inductive proximity switch.(1 u) *Check its mounting, clean the sensing face, and ensure alignment with the metallic activator on the plate *Inspect wiring condition. CLEANING *Clean around the TLE Profile assembly (shaft, washers, pivot area).
